Gradiente de un campo escalar

$$\underline{u} = \nabla \Phi \quad \Rightarrow \quad u_i = \frac{\partial \Phi}{\partial x_i}$$

$$\underline{u} = \begin{bmatrix} \frac{\partial \Phi}{\partial x_1} \\ \frac{\partial \Phi}{\partial x_2} \\ \frac{\partial \Phi}{\partial x_3} \end{bmatrix} = \begin{bmatrix} \frac{\partial \Phi}{\partial x} \\ \frac{\partial \Phi}{\partial y} \\ \frac{\partial \Phi}{\partial z} \end{bmatrix}$$

Gradiente de un campo vectorial (tensor de orden uno)

$$\underline{\underline{U}} = \underline{\nabla} \underline{v} \quad \Rightarrow \quad U_{ij} = \frac{\partial v_i}{\partial x_j}$$

$$\underline{\underline{U}} = \begin{bmatrix} \underline{\nabla} v_x^T \\ \underline{\nabla} v_y^T \\ \underline{\nabla} v_z^T \end{bmatrix} = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial y} \end{bmatrix}$$

Divergencia de un campo vectorial (tensor de orden uno)

$$u = \underline{\nabla} \cdot \underline{v} = \frac{\partial v_i}{\partial x_i}$$

$$u = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

Divergencia de un campo tensorial de orden 2:

$$\underline{u} = \underline{\nabla} \cdot \underline{\underline{T}} \quad \Rightarrow \quad u_i = \frac{\partial T_{ij}}{\partial x_j}$$

$$\underline{u} = \begin{bmatrix} \frac{\partial T_{xx}}{\partial x} + \frac{\partial T_{xy}}{\partial y} + \frac{\partial T_{xz}}{\partial z} \\ \frac{\partial T_{yx}}{\partial x} + \frac{\partial T_{yy}}{\partial y} + \frac{\partial T_{yz}}{\partial z} \\ \frac{\partial T_{zx}}{\partial x} + \frac{\partial T_{zy}}{\partial y} + \frac{\partial T_{zz}}{\partial z} \end{bmatrix}$$

Rotor de un campo vectorial (tensor de orden uno):

$$\underline{u} = \underline{\nabla} \times \underline{v} \Rightarrow u_i = \epsilon_{ijk} \cdot \frac{\partial v_k}{\partial x_j} = \epsilon_{ijk} \cdot \overset{u_j}{\frac{\partial}{\partial x_j}} \overset{v_k}{\left(v_k \right)}$$

Forma vectorial de escribir rotor: $\underline{u} = \epsilon_{ijk} \cdot \frac{\partial v_k}{\partial x_j} \cdot \underline{e}_i$

$$\underline{u} = \left(\overset{123}{\frac{\partial v_3}{\partial x_2}} - \overset{132}{\frac{\partial v_2}{\partial x_3}} \right) \underline{e}_1 + \left(\overset{231}{\frac{\partial v_1}{\partial x_3}} - \overset{213}{\frac{\partial v_3}{\partial x_1}} \right) \underline{e}_2 + \left(\overset{312}{\frac{\partial v_2}{\partial x_1}} - \overset{321}{\frac{\partial v_1}{\partial x_2}} \right) \underline{e}_3$$

determinante

$$\underline{u} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$$

Laplaciano de un campo escalar:

$$u = \overbrace{\underline{\nabla} \cdot \underline{\nabla}}^{\nabla^2 \equiv \Delta} \Phi = \frac{\partial}{\partial x_i} \left(\frac{\partial \Phi}{\partial x_i} \right) = \frac{\partial^2 \Phi}{\partial x_i \partial x_i}$$

$$u = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2}$$

Laplaciano de un campo vectorial:

$$\underline{u} = \underline{\nabla} \cdot \underline{\nabla} \underline{v} \Rightarrow u_i = \frac{\partial}{\partial x_j} \left(\frac{\partial v_i}{\partial x_j} \right) = \frac{\partial^2 v_i}{\partial x_j \partial x_j}$$

$$\underline{u} = \begin{bmatrix} \underline{\nabla} \cdot \underline{\nabla} v_x \\ \underline{\nabla} \cdot \underline{\nabla} v_y \\ \underline{\nabla} \cdot \underline{\nabla} v_z \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_i} \left(\frac{\partial v_x}{\partial x_i} \right) \\ \frac{\partial}{\partial x_i} \left(\frac{\partial v_y}{\partial x_i} \right) \\ \frac{\partial}{\partial x_i} \left(\frac{\partial v_z}{\partial x_i} \right) \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \\ \frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \\ \frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \end{bmatrix}$$